

AI Acceptance in the Sharing Economy

Data Collection and Methodology — Working Report

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1 The idea

The **sharing economy** — peer-to-peer marketplaces where private owners rent out homes (Airbnb, Vrbo), cars and RVs (Turo, Getaround, RVshare, Outdoorsy), and creative equipment such as cameras and drones (Fat Llama, KitSplit, Lensrentals) — runs entirely on **trust between strangers**. A host has to trust a guest with their property; a renter has to trust that a listing is real and a counterpart is fair. These platforms increasingly insert **AI and AI agents** into exactly that trust relationship: automated guest screening, chatbot support, dynamic (“smart”) pricing, automated messaging, and damage-assessment models.

Our research question is therefore:

How do users perceive AI and AI agents in peer-to-peer rental platforms, and what drives or erodes their trust — across different rental types and across two very different kinds of text: open community discussion (Reddit) versus paying-customer reviews (Trustpilot)?

We deliberately collect **two complementary corpora**. Following the professor’s framing, **Reddit is our baseline** — the place where users *deliberate*, argue, complain, and speculate about AI, hosting, and rentals in their own words. **Trustpilot is our verification layer** — structured post-transaction reviews that come with a **1–5 star rating that acts as a ground-truth sentiment label**, with no lexicon required. The two are not identical sources, and that is the point: where the open Reddit discussion and the labelled Trustpilot reviews *agree*, we gain confidence; where they *diverge*, we learn something about the gap between public discourse and actual customer experience.

The remainder of this report documents (i) how each corpus was collected, targeted, balanced and cleaned; (ii) what the resulting datasets look like; and (iii) the analysis steps each team member will carry out next.

2 Data source 1 — Reddit (the baseline)

We collected Reddit data with the `RedditExtractor` package, which queries Reddit’s public API. The corpus went through **three passes**; each fixed a problem in the previous one.

Table 1: Three Reddit collection passes. Output files, in order: `reddit_all.csv`, `reddit_targeted.csv`, `reddit_clean.csv`.

Pass	Script	Rows	Status
v1 – keyword (global)	<code>scraping_data_v1.R</code>	14,090	baseline only (off-topic)
v2 – subreddit-targeted	<code>01_scrape_reddit.R</code>	1,009	raw, on-topic
v2 – cleaned + themed	<code>05_clean_reddit.R</code>	314	USED for analysis

2.1 Pass 1 — why the first scrape failed

The first attempt searched **all of Reddit** with a single bag of generic keywords ("AI rental platform helpful automation trust"). This returned a large but essentially **off-topic** corpus of roughly 14,090 comments: threads about K-pop drama, gaming, *Am I the Asshole* stories and US politics dominated, and **fewer than 1 % of comments even mentioned “rental”**. We keep this file (`data/reddit_all.csv`) only as a transparency baseline — it is the “before” picture that motivated the rebuild.

2.2 Pass 2 — how we *targeted* the scrape

The fix was to stop searching all of Reddit and instead search **inside topic-relevant subreddits**. `RedditExtractor::find_thread_urls()` lets us restrict a keyword search to a single subreddit, which is what makes the results on-topic. We defined **47 (subreddit, keyword) search pairs** organised in six thematic layers:

- **Home / stay sharing** — `r/AirBnB`, `r/airbnb_hosts`, `r/AirBnBHosts`, `r/vrbo`, `r/ShortTermRentals` · *keywords*: “AI”, “host AI”, “automated message”, “co-host”
- **Car / RV sharing** — `r/Turo`, `r/gigwork`, `r/RVing`, `r/GoRVing` · *keywords*: “AI”, “automated”, “trust”, “rental AI”
- **Camera / film / drone gear** — `r/photography`, `r/AskPhotography`, `r/cinematography`, `r/videography`, `r/Filmmakers`, `r/drones`, `r/dronephotography`, `r/DJI` · *keywords*: “AI rental”, “lens rental”, “rent gear”
- **Event tech** — `r/eventproduction`, `r/EventProduction`, `r/livesound` · *keywords*: “AI”
- **AI agents in commerce** — `r/Entrepreneur`, `r/SaaS`, `r/startups`, `r/automation`, `r/ChatGPT`, `r/OpenAI`, `r/artificial`, `r/singularity` · *keywords*: “AI agent”, “automation rental”, “agent booking”
- **Trust / privacy** — `r/privacy`, `r/technology` · *keywords*: “AI agent”, “AI assistant data”, “AI agent trust”

The scrape parameters were:

- `threads_per_search = 30` — up to 30 most relevant threads per pair.
- `sort_by = "relevance"`, `period = "all"` — full Reddit history, so older threads that have accumulated discussion are included.
- `comments >= 6` — a thread is only fetched if it has real discussion; 1–2 reply threads are dropped.

- **max_threads = 400** — a ceiling on total threads fetched to keep the rate-limited comment pull to a reasonable wall-clock.
- Comments shorter than 20 characters, `[deleted]`/`[removed]` bodies, and duplicate comment IDs are discarded.

Crucially, **the keyword used to *find* a thread does not become its label**. Keywords only locate relevant discussion; sentiment and topic are derived later from the actual comment text. This keeps the scrape from biasing the analysis toward whatever words we happened to search for.

This pass produced **1,009 on-topic comments** (`data/reddit_targeted.csv`).

2.3 Pass 3 — how we *balanced* and cleaned

Two mechanisms keep a handful of viral threads from dominating the corpus:

1. **Per-thread comment cap (in `01_scrape_reddit.R`)**. The script can write a *balanced* version that caps each thread at **120 comments**, keeping the highest-upvoted ones, so one 500-comment thread cannot swamp the dataset.
2. **Dropping the noisy subreddits (`05_clean_reddit.R`)**. Even after capping, two subreddits — **r/Filmmakers** and **r/Entrepreneur** — still dominated, because they hosted three viral threads about AI replacing humans in *creative and business careers* (e.g. “Sir Roger Deakins”, “I don’t think these guys actually like movies”). Those debates are off-topic for *rental-platform* trust, so we remove both subreddits entirely.

The cleaning step also tags every remaining comment with a **theme** column:

- **sharing_economy** — rental, host, car/RV and equipment-gear subreddits;
- **ai_tech** — AI, agent, privacy and technology subreddits.

The result is `data/reddit_clean.csv`, the file all downstream analysis uses.

2.4 The Reddit dataset

After cleaning, the corpus is **314 comments** drawn from **17 subreddits** and **38 distinct threads**, spanning **Oct 2012 – May 2026**, with a mean comment length of **239 characters**. The two themes are close to evenly balanced:

Table 2: Reddit theme balance (`reddit_clean.csv`).

Theme	Comments	Share
ai_tech	146	46%
sharing_economy	168	54%

Table 3: Top subreddits in the cleaned Reddit corpus.

Subreddit	Theme	Comments
ChatGPT	ai_tech	81
photography	sharing_economy	76
privacy	ai_tech	56
GoRVing	sharing_economy	38

Subreddit	Theme	Comments
AirBnB	sharing_economy	22
dji	sharing_economy	13
turo	sharing_economy	9
artificial	ai_tech	5
cinematography	sharing_economy	5
technology	ai_tech	2
AirBnBHosts	sharing_economy	1
EventProduction	sharing_economy	1

3 Data source 2 — Trustpilot (the verification layer)

3.1 Why Trustpilot

Trustpilot reviews give us something Reddit cannot: a **structured 1–5 star rating attached to every piece of text**. That star rating is a **ground-truth sentiment label** we did not have to infer. This is exactly what makes Trustpilot a *verification* layer — we can compute a sentiment score on the review text and check it against the star the reviewer actually gave, then carry that validated lexicon over to the unlabelled Reddit comments (convergent validity).

3.2 How we scraped it

Trustpilot blocks naive scraping, so we used the **Apify** actor `casper11515/trustpilot-reviews-scraper` through the **Apify web UI**, running it **once per company**. For each platform we entered its Trustpilot company page, capped the run at roughly 1,000–2,000 reviews, and downloaded the result CSV into `data/trustpilot_raw/`. The script `02_merge_trustpilot.R` then:

- reads every CSV in that folder (robust to BOM / carriage-return / invalid-UTF-8 issues from the export);
- normalises the actor’s columns into a canonical schema (`platform`, `rating`, `title`, `body`, `author`, `country`, `date_posted`, ...);
- keeps English reviews with at least 20 characters of text and drops duplicate review IDs.

The merged output is `data/trustpilot_reviews.csv`.

3.3 The Trustpilot dataset

The merged file holds **3,912 unique English reviews** across **7 platforms**, spanning **Jul 2019 – May 2026**. Each row carries a 1–5 star rating, review title and body, reviewer country, and the company’s reply where one exists.

Table 4: Trustpilot reviews per platform, with mean star rating.

Platform	Reviews	Mean rating	Date range
Turo	1482	4.62	May 2026 – May 2026
Fat Llama	835	4.11	Jun 2020 – Jan 2026
RVshare	773	3.98	Sep 2023 – May 2026
Outdoorsy	751	3.68	Aug 2023 – May 2026
Lensrentals	35	2.94	Dec 2020 – May 2026
Getaround	31	4.35	May 2026 – May 2026
KitSplit	5	1.00	Jul 2019 – Jun 2023

Table 5: Trustpilot star-rating distribution (the ground-truth labels).

Stars	Reviews	Share
1	593	15%
2	113	3%

Stars	Reviews	Share
3	139	4%
4	213	5%
5	2854	73%

Collapsed into classes, the reviews split as **706 negative (1-2 stars)**, **139 neutral (3 stars)**, and **3067 positive (4-5 stars)**. The strong positive skew is typical of opt-in review platforms and means we will **down-sample the positive class** before any balanced classification.

4 How the two sources fit together

Table 6: Reddit (baseline) vs Trustpilot (verification).

Dimension	Reddit	Trustpilot
Role	Baseline	Verification
Text type	Open discussion	Post-transaction review
Sentiment label	Inferred (AFINN / NRC)	Ground truth (1-5 stars)
Volume	314	3,912
Time span	Oct 2012 – May 2026	Jul 2019 – May 2026

Reddit answers *what users talk about* around AI and rentals; Trustpilot answers *how paying customers rate* the experience. The triangulation logic is: **(1)** compute lexicon sentiment on Trustpilot review text; **(2)** validate it against the star rating (do 1-star reviews score negative?); **(3)** apply the *validated* lexicon to Reddit, where no labels exist; **(4)** compare the *topics* that drive negative sentiment in each source. Agreement across two structurally different corpora is far stronger evidence than either source alone.

5 Next steps

Note. The paragraphs below are working drafts / placeholders. Each owner will expand their section with the actual figures, tables and interpretation once the analysis scripts are run. Assignments are provisional — swap as suits the team’s strengths.

5.1 Preprocessing and topic modelling — Htet Phone Khaing

Draft. This section will document the shared preprocessing pipeline (`06_preprocess.R`): lower-casing, removal of URLs and Reddit/Markdown artefacts, stop-word removal, lemmatisation, and tokenisation into both unigrams and bigrams for the two corpora. It will then present the topic model (`07_lda.R`): we fit 3–4 LDA topics per theme (the corpus is small — 38 Reddit threads — so we cap at four topics to avoid over-fragmentation), report the top terms per topic, and run a seeded LDA using trust-related seed words (`trust`, `safe`, `scam`, `verified`, `automated`) to test whether a coherent “trust” topic emerges. Expected finding to confirm or revise: AI-side topics lean toward “AI tools / plugins” because one r/ChatGPT thread is large, while sharing-economy topics separate into hosting logistics, pricing, and gear-quality clusters.

5.2 Visualisation and reporting — Tatum Miranda Smith

Draft. This section will hold the required visual analysis built on top of the preprocessed tokens: top-10 word-frequency bar charts per theme; comparison and commonality word clouds (unigram and bigram); word co-occurrence networks (`widyr::pairwise_count + ggraph`); and a single sentiment time-series overlaying the two sources. It will also fold these figures, plus the dataset tables from this report, into the final paper. Draft narrative to complete: “car- and RV-sharing platforms cluster at 3.7–4.6 stars on Trustpilot while equipment rental sits far lower, and the Reddit discussion mirrors this split in its vocabulary of complaints.”

5.3 Sentiment, embeddings and synthesis — Artem Cherkaskyy

Draft. This section will compute AFINN and NRC sentiment on both corpora, validate the lexicon against Trustpilot stars (the convergent-validity check), and report the NRC eight-emotion profile per source. It will add GloVe word embeddings (`text2vec`) to inspect the nearest neighbours of “trust”, “AI” and “host”, then map the findings onto established trust frameworks (Mayer et al. 1995 — ability / benevolence / integrity; TAM). The section closes with the project’s practical contribution: a set of **AI-agent design cards** — concrete, data-backed recommendations for how a rental platform should (and should not) deploy AI in the trust relationship.

6 Limitations

- **Trustpilot skews positive** (opt-in bias) — we down-sample for balanced classification.
- **Equipment-rental platforms are thin** (Lensrentals and KitSplit have few Trustpilot reviews), so per-platform claims there are weak.
- **One Reddit thread is large** relative to the cleaned corpus, pulling the AI-side topics toward “AI tools” language; we note this rather than hide it.
- **Few Reddit threads overall** (38) — the realistic LDA ceiling is 3–4 topics per theme.
- **“Trust” is mostly implicit** in both corpora (expressed through host/renter, payment and review language) rather than as the literal word, so we measure it via topics and sentiment, not keyword counts.
- **English-only**, and Reddit’s anti-bot wall now prevents fresh anonymous re-scrapes (reproduction requires an authenticated OAuth app — see `legacy/scrape_reddit_oauth_attempt.R`).
- **Per-platform review dates need a parse check:** for Turo and Getaround the computed date range collapses to a single month, which points to a date-format inconsistency in those two Apify exports. The overall span is reliable, but `02_merge_trustpilot.R` should be revisited before any sentiment-over-time analysis that splits by platform.

Repository: <https://github.com/tyomachkaa/ai-acceptance-sharing-economy> · *Course:* Online Content Analysis, WU Vienna, 2026.